

The anthropomorphism fallacy misinterprets LLM responses, treating statistical artifacts as expressions of an inner mental life. When researchers ascribe human-like mental states—such as beliefs, intentions, or consciousness—to LLMs based on their linguistic output, they risk invalidating measures of psychological constructs. Indeed, many such constructs, including attitudes and emotions, presuppose a mental architecture that current LLMs lack.

In many situations, using anthropomorphic language in conversation is natural—even useful. However, with AI like LLMs—which currently lack documented markers of consciousness yet produce remarkably coherent conversations (Shardlow & Przybyla, 2024)—this semblance seduces users to interpret model behavior through the lens of folk psychology, attributing “beliefs” or “consciousness” to these systems (Colombatto & Fleming, 2024). Indeed, the chatbot service provider Character AI invites users to meet AIs that “feel alive.”

While the philosophical debate about machine consciousness remains open (Butlin et al., 2023), ascribing emotions or intentions to current LLMs (saying they “believe” or “think”) risks creating impressions that outpace their demonstrated capabilities. As of mid-2025, no publicly available LLM exhibits clear markers of phenomenal awareness or intentional agency. Such attribution gaps can lead researchers and the public to either overestimate or underestimate these systems’ capabilities (Crockett & Messeri, 2023; Shanahan, 2024), potentially shaping AI policy in ways disconnected from technological reality (Lin, 2025). Perhaps most concerning, this conflation may dilute our understanding of distinctly human qualities—feelings, thoughts, and virtues—thereby diminishing their meaning (Vallor, 2024).

The fallacy also leads researchers to draw invalid inferences from LLM outputs. When interpreting model responses as expressing human-like mental states rather than statistical probabilities, they may fail to account for the statistical artifacts inherent in token prediction systems. Similarly, findings may be inappropriately generalized from anthropomorphized LLMs to human populations without recognizing the essential difference between statistical text generation and human psychological processes.

The replacement perspective thus risks anthropomorphizing algorithms and mischaracterizing their nature—a conceptual error that invites misunderstandings and misinterpretations. For example, one such mischief is that “any given LLM can act as only a single participant” (Dillion et al., 2023). Yet unlike humans, who are influenced by a unique combination of personal experiences, emotions, and cognitive biases, LLMs are not limited to a single perspective but generate responses based on their vast, diverse dataset. This means that, depending on the prompt and context, the same LLM can produce a range of patchwork responses, each reflecting different viewpoints or types of reasoning (Santurkar et al., 2023). This variability is not indicative of a singular, consistent “mind,” but rather of a multifaceted tool capable of simulating diverse perspectives. A teenager, a senior citizen, a subject matter expert, or a layperson: LLMs can role-play various characters or personas (Shanahan et al., 2023). This chameleon-like ability highlights LLMs as tools for linguistic simulation, not as human participants.

Mitigating the anthropomorphism fallacy requires using precise language, maintaining conceptual clarity about LLMs’ lack of mental states, documenting technical settings, applying simulation-based interpretive frameworks, and specific researcher training. Consider:

1. **Language and framing:** Use precise, non-anthropomorphic terminology when describing LLM outputs. Instead of saying an LLM “believes” or “feels,” opt for terms like “produces,” “generates,” or “outputs” to accurately reflect their statistical nature (Ibrahim & Cheng, 2025; Shanahan, 2024).

2. **Conceptual clarity:** Explicitly acknowledge in research designs and reports that current LLMs do not possess mental states or consciousness (Shardlow & Przybyla, 2024). Define psychological constructs carefully, noting when they inherently depend on mental states that LLMs lack.
3. **Documentation practices:** Document the specific LLM, version, prompt design, and parameter settings used to generate responses (Lin, 2025), emphasizing the technical rather than psychological aspects of the process.
4. **Interpretive frameworks:** Develop and apply interpretive frameworks that treat LLM outputs as simulations rather than expressions of beliefs or attitudes (Ibrahim & Cheng, 2025). This includes distinguishing between “simulated beliefs” and actual beliefs when reporting results.
5. **Education and training:** Provide training to research teams on the mechanisms of LLMs and the risks of anthropomorphic interpretations (Lin, 2025). Foster a research culture that maintains conceptual precision when discussing AI capabilities.

### Identity Essentialization Fallacy

Under the replacement perspective, prompting often invokes identity labeling, such as instructing the LLM to act as or adopt the identity of “White man,” “Black woman,” “Chinese,” or “American”—as if such labels describe innate, static, homogeneous social groups, each entailing a specific set of behaviors (Chuang et al., 2024; Wang et al., 2025). This approach constitutes an identity essentialization fallacy that caricatures how identity operates in human populations.

When simple demographic labels are used to prompt LLMs, there is often an implicit assumption that these can generate responses representative of real human populations. Such prompting approaches treat social categories as static and homogeneous, ignoring the vast diversity within any demographic group. When LLMs generate responses based on these simplified identity prompts, they may produce stereotyped or inaccurate outputs that fail to capture the nuanced realities of actual human populations (Li & Qi, 2025), limiting what we can learn about real-world contexts and populations (Lahoti et al., 2023; M. H. Lee et al., 2024). Indeed, demographic prompting can even reduce alignment with human judgments (Sun et al., 2025).

In colloquial exchanges, essentialist language about social categories—from “artists are eccentric” to “women are nurturing”—is convenient and also meaningful. But in empirical research, identity essentialization masks the fluidity and diversity inherent within any demographic, overlooking individual nuances and intersectionality while reinforcing stereotypes and biases prevalent within society, thus overestimating group differences (Namboodiripad et al., 2023; Prentice & Miller, 2006).

Identity is not a static, unitary construct that can be captured by a single demographic label—it is fluid, contextual, and intersectional. When researchers use essentialist prompting techniques, they misrepresent the psychological construct of identity itself, reducing rich, complex human experiences to one-dimensional categories. This reductive operationalization fails to capture how various aspects of identity interact, how identity salience shifts across contexts, and how individuals negotiate multiple, sometimes contradictory, identity facets.

This is not to deny the importance of identities, nor to advocate for identity-blindness. As pervasive societal structures that shape our thoughts, attitudes, and behaviors, social categories like race, gender, and class are deeply embedded in our experiences—and often an ingrained part of our identity. But rather than reducing individuals to essentialist categories, a more appropriate approach is to consider how various identities—demographic, professional, or situational—interact, by role-playing various personas through contextualized prompting. This involves crafting character profiles that encompass a broader array of characteristics—

from contextual descriptions (“I am a young tech worker living in the United States”) to broader social categories (e.g., based on political leaning or personality type)—allowing for more nuanced explorations of perspectives and experiences.

For example, instead of prompting an LLM to act as a “Black woman,” which may reinforce stereotypes or oversimplify identity (Sun et al., 2025), we might construct a more holistic persona by adding a specific context, such as “a young entrepreneur from Atlanta who is passionate about sustainable fashion and community development”; or by incorporating intersectional identities, such as “a young Black female tech worker navigating the challenges of a male-dominated field.” These contextual descriptions incorporate identity but frame it within specific experiences, values, and contexts. Indeed, contextualized prompting has been shown to evoke distinct, diverse (Wang et al., 2025), and better-aligned responses from LLMs (Bui et al., 2025).

Identity is multifaceted and context-dependent, with varying salience for different individuals. Simulating human participants therefore risks misrepresenting the salience of various aspects of identity—reflecting the prompter’s perspective or presumptions about which aspects of identity are important, rather than capturing the intersectional reality experienced by the simulated persona. It is therefore crucial, whether using contextualized prompting or not, to examine potential biases and limitations in the prompt. Sidestepping genuine engagement with marginalized communities further risks artificial inclusion (Agnew et al., 2024).

To mitigate the identity essentialization fallacy when simulating diverse perspectives, safeguards include using contextual prompting, incorporating intersectional approaches, performing diversity validation, ensuring transparency about limitations, and adopting collaborative methods. Consider several approaches:

1. **Contextual prompting:** Instead of using simple demographic labels, develop richer, context-specific prompts that incorporate multiple aspects of identity (including intersectionality), specific experiences, and environmental factors (Bui et al., 2025).
2. **Validation:** Compare LLM-generated responses across multiple prompting strategies and validate against actual human responses from the target population to identify when simulations misrepresent or stereotype particular groups (Sun et al., 2025).
3. **Transparency in limitations:** Explicitly acknowledge the limitations of identity simulation in research reports, including the risk of reinforcing stereotypes or oversimplifying complex identities. Document the specific prompting approaches used and their potential biases (Sun et al., 2025).
4. **Collaborative approach:** When studying specific cultural or identity groups, involve members of those groups in designing prompts, validating outputs, and interpreting results (Zhao et al., 2024).

### Substitution Fallacy

LLMs can mimic certain aspects of human behavior and cognition, but purporting them as primary tools to directly reveal the human mind reflects a substitution fallacy.

A core issue arises from the temporal limitations of LLM training data. Because LLMs are trained on historical datasets with a specific cutoff date, they represent a snapshot of human knowledge, attitudes, and behaviors at that moment. Updating through retraining is infrequent and resource-intensive. This static nature restricts their capacity to capture ongoing societal changes, new social phenomena, or evolving attitudes and behaviors, such as rapidly changing views on new technologies or social movements (Zhu et al., 2025). Without real-time adaptability, previous alignments do not guarantee current applicability. Furthermore, a model’s advertised knowledge cutoff often differs from its actual, or effective, knowledge cutoff. The functional knowledge of LLMs frequently corresponds to older text